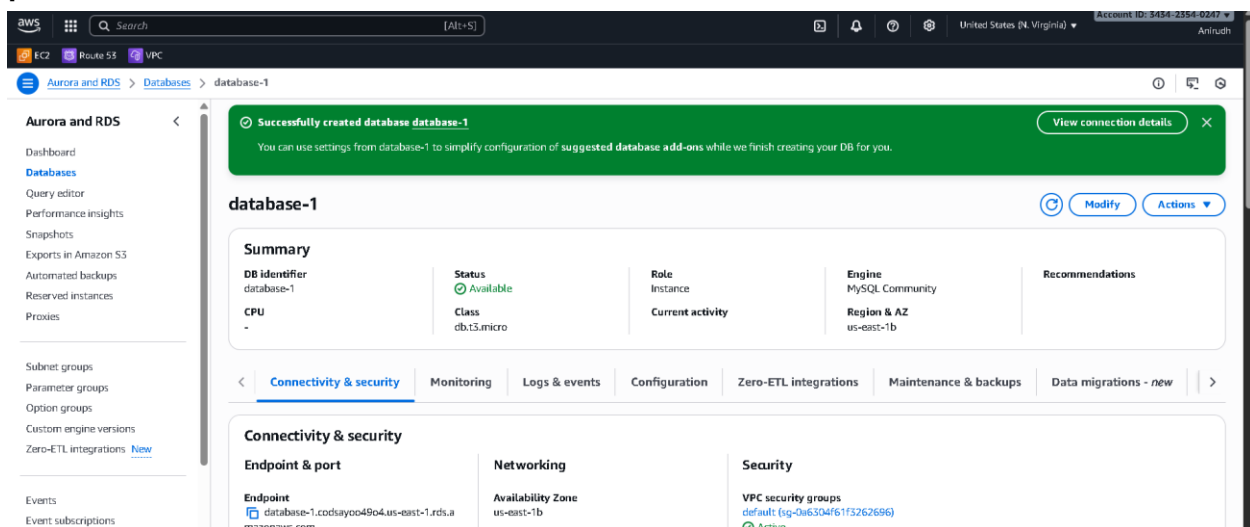


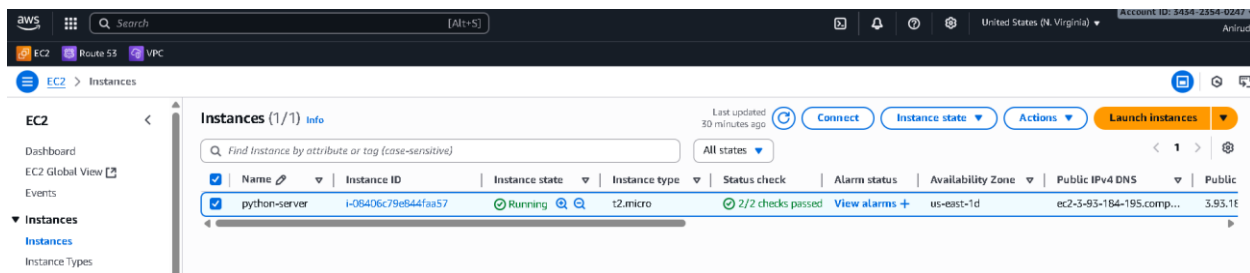
Today's Task 22-09-2025

Task : To connect a python backend application to an RDS database.

First create the RDS Database. And make the Database private.



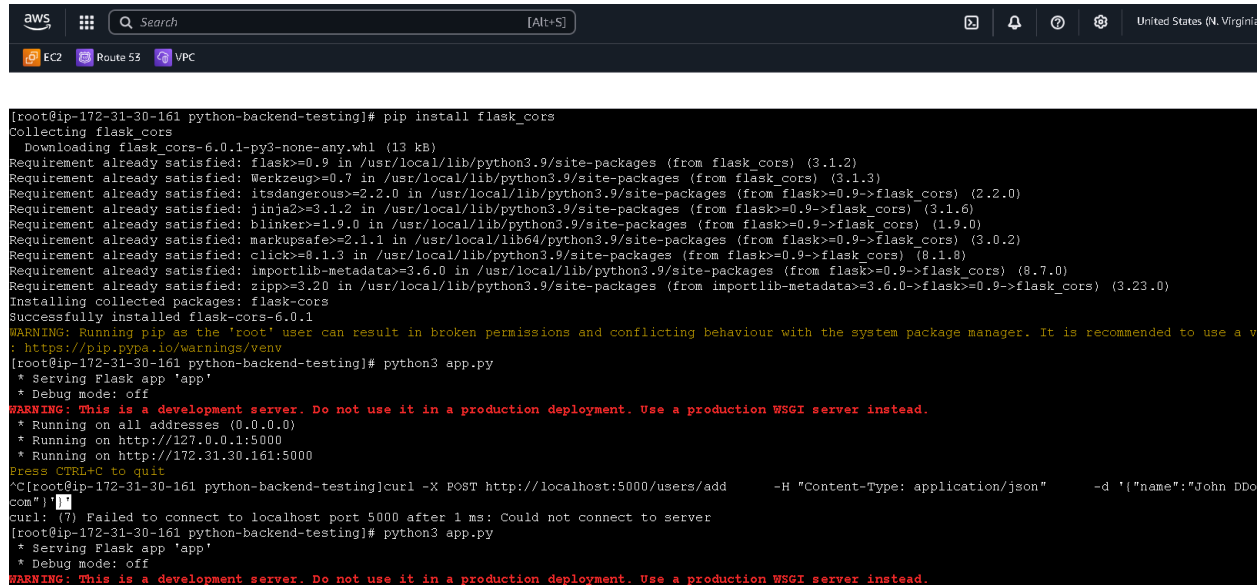
Now create an EC2 server and connect to the SSH .



In SSH , install git by

yum install git -y


```
yum install python3-pip -y
pip install flask mysql-connector-python
pip install flask_cors
```



```
aws
[Alt+S]
EC2 Route 53 VPC

[root@ip-172-31-30-161 python-backend-testing]# pip install flask_cors
Collecting flask_cors
  Downloading flask_cors-6.0.1-py3-none-any.whl (13 kB)
Requirement already satisfied: flask<=0.9 in /usr/local/lib/python3.9/site-packages (from flask_cors) (3.1.2)
Requirement already satisfied: Werkzeug<=0.7 in /usr/local/lib/python3.9/site-packages (from flask_cors) (3.1.3)
Requirement already satisfied: itsdangerous<=2.2.0 in /usr/local/lib/python3.9/site-packages (from flask_cors) (2.2.0)
Requirement already satisfied: Jinja2<=3.1.2 in /usr/local/lib/python3.9/site-packages (from flask_cors) (3.1.6)
Requirement already satisfied: blinker<=1.9.0 in /usr/local/lib/python3.9/site-packages (from flask_cors) (1.9.0)
Requirement already satisfied: MarkupSafe<=2.1.1 in /usr/local/lib/python3.9/site-packages (from flask_cors) (3.0.2)
Requirement already satisfied: click<=8.1.3 in /usr/local/lib/python3.9/site-packages (from flask_cors) (8.1.8)
Requirement already satisfied: importlib-metadata<=3.6.0 in /usr/local/lib/python3.9/site-packages (from flask_cors) (8.7.0)
Requirement already satisfied: zipp<=3.20 in /usr/local/lib/python3.9/site-packages (from importlib-metadata==3.6.0->flask_cors) (3.23.0)
Installing collected packages: flask-cors
Successfully installed flask-cors-6.0.1
WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour with the system package manager. It is recommended to use a v
: https://pip.pypa.io/warnings/venv
[root@ip-172-31-30-161 python-backend-testing]# python3 app.py
* Serving Flask app 'app'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.31.30.161:5000
Press CTRL+C to quit
^C[root@ip-172-31-30-161 python-backend-testing]# curl -X POST http://localhost:5000/users/add -H "Content-Type: application/json" -d '{"name": "John Doe"}'
curl: (7) Failed to connect to localhost port 5000 after 1 ms: Could not connect to server
[root@ip-172-31-30-161 python-backend-testing]# python3 app.py
* Serving Flask app 'app'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
```

Then run the python application by

```
python3 app.py
```

Now duplicate the SSH server , and keep running the application in the first SSH instance.

In the second SSH instance , install mysql by the following commands.

As we created the DB instance in private , we have to use it with EC2.

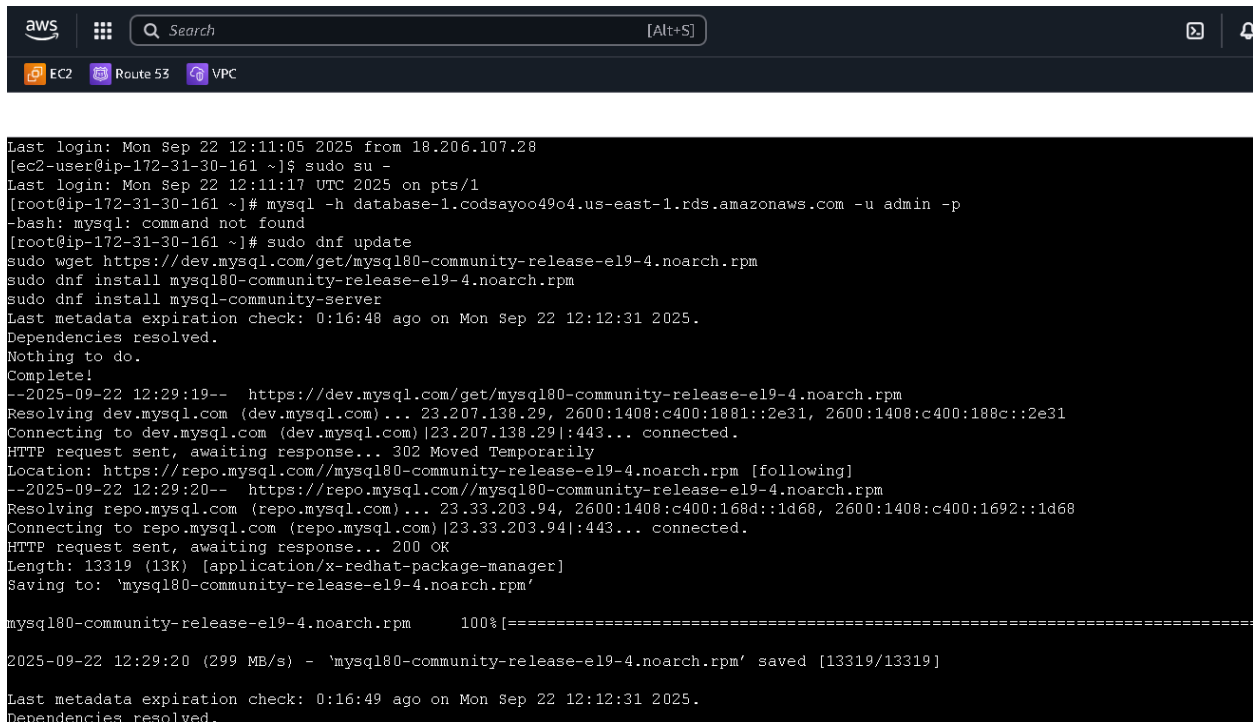
Anirudh Trivedi

sudo dnf update

sudo wget https://dev.mysql.com/get/mysql80-community-release-el9-4.noarch.rpm

sudo dnf install mysql80-community-release-el9-4.noarch.rpm

sudo dnf install mysql-community-server

A screenshot of an AWS Management Console terminal window. The terminal shows the following commands and output:

```
Last login: Mon Sep 22 12:11:05 2025 from 18.206.107.28
[ec2-user@ip-172-31-30-161 ~]$ sudo su -
Last login: Mon Sep 22 12:11:17 UTC 2025 on pts/1
[root@ip-172-31-30-161 ~]# mysql -h database-1.codsayoo49o4.us-east-1.rds.amazonaws.com -u admin -p
-bash: mysql: command not found
[root@ip-172-31-30-161 ~]# sudo dnf update
sudo wget https://dev.mysql.com/get/mysql80-community-release-el9-4.noarch.rpm
sudo dnf install mysql80-community-release-el9-4.noarch.rpm
sudo dnf install mysql-community-server
Last metadata expiration check: 0:16:48 ago on Mon Sep 22 12:12:31 2025.
Dependencies resolved.
Nothing to do.
Complete!
--2025-09-22 12:29:19-- https://dev.mysql.com/get/mysql80-community-release-el9-4.noarch.rpm
Resolving dev.mysql.com (dev.mysql.com)... 23.207.138.29, 2600:1408:c400:1881::2e31, 2600:1408:c400:188c::2e31
Connecting to dev.mysql.com (dev.mysql.com)|23.207.138.29|:443... connected.
HTTP request sent, awaiting response... 302 Moved Temporarily
Location: https://repo.mysql.com//mysql80-community-release-el9-4.noarch.rpm [following]
--2025-09-22 12:29:20-- https://repo.mysql.com//mysql80-community-release-el9-4.noarch.rpm
Resolving repo.mysql.com (repo.mysql.com)... 23.33.203.94, 2600:1408:c400:168d::1d68, 2600:1408:c400:1692::1d68
Connecting to repo.mysql.com (repo.mysql.com)|23.33.203.94|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 13319 (13K) [application/x-redhat-package-manager]
Saving to: 'mysql80-community-release-el9-4.noarch.rpm'

mysql80-community-release-el9-4.noarch.rpm    100%[=====]
2025-09-22 12:29:20 (299 MB/s) - 'mysql80-community-release-el9-4.noarch.rpm' saved [13319/13319]

Last metadata expiration check: 0:16:49 ago on Mon Sep 22 12:12:31 2025.
Dependencies resolved.
```

Now connect the RDS to EC2 by the following command.

mysql -h <rds_endpoint> -u <admin> -p

Enter the password.

Now you have connected to RDS.

Now create the database and table in the mysql .

Anirudh Trivedi

```
CREATE DATABASE dev;  
USE dev;
```

```
CREATE TABLE users (  
    id INT AUTO_INCREMENT PRIMARY KEY,  
    name VARCHAR(100) NOT NULL,  
    email VARCHAR(100) NOT NULL UNIQUE  
);
```

Now we can select , insert and delete the rows and elements in the table by the following curl commands.

For GET :

```
curl -X GET http://localhost:5000/users
```

For POST :

```
curl -X POST http://localhost:5000/users/add -H "Content-Type: application/json" -d '{"name": "John DDoe", "email": "john@example.com"}'
```

For GET : (for single user)

```
curl -X GET http://localhost:5000/users/1
```

For PUT :

```
curl -X PUT http://localhost:5000/users/update/1 \  
    -H "Content-Type: application/json" \  
    -d '{"name": "John Updated", "email": "  
    "john.updated@example.com"}'
```

For DELETE :

```
curl -X DELETE http://localhost:5000/users/delete/1
```

